

Threat Intelligence Ecosystem for Banking



Chaima Memi



Ahmed Maalej



Molka Dhrief



Mohamed Gassoumi



Sinda Galai



Moslem Haddedi



Nadhem Laameri

Secure7

4 NIDS1



Table of Contents

Introduction & Context	1	Problem & Business Need	2
Project Objectives & Deliverables	3	Solution Overview	4
Methodology & Implementation Plan	5	Architecture & Technical Highlights	6
Resource & Infrastructure Requirements	7	Risks, Compliance & Assumptions	8
Cost Breakdown & ROI	9	KPIs & Performance Metrics	10
Results & Business Value	11	Conclusion & Next Steps	12

1. Introduction & Context



A close-up, slightly blurred photograph of a person's hand typing on a laptop keyboard. The image is overlaid with a semi-transparent blue filter. Centered over the image is the text "It's not just **viruses**, but **people**." in a white, sans-serif font. The words "viruses" and "people" are highlighted in a light blue color.

It's not just **viruses**, but **people**.

2. Problem & Business Need



2. Problem & Business Need

Core Problem

- Lack of real-time visibility into internal systems
- Delayed detection of intrusions
- Manual, slow response processes
- Traditional tools are siloed and reactive



2. Problem & Business Need

Business Need

- Detects threats in real time
- Traps attackers early (via deception)
- Automates and accelerates incident response
- Helps ensure legal and regulatory compliance



3. Project Objectives & Deliverables





**Enhance Cybersecurity
Posture**



**Achieve Regulatory
Compliance**

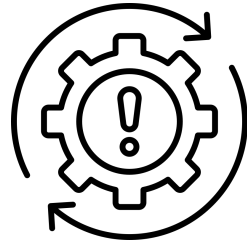


**Reduce
Operational Costs**



**Protect Customer
Trust and Data**

**Business
Goals**

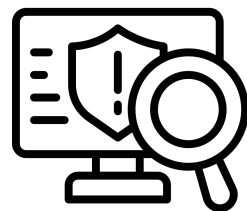


**Automated Incident
Response**

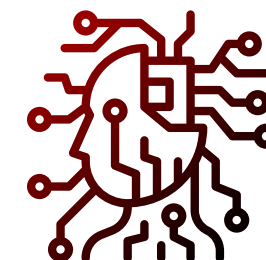


**Endpoint and Network
Security Coverage**

Technical Goals



**Real-Time Threat
Detection and Correlation**



**Threat Intelligence
Enrichment**

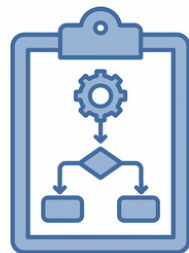


**Integrated Threat
Intelligence Platform**



**Performance Testing
and Validation Reports**

Key Delivrables



**Automated Incident
Response Playbooks**



**Security Policies
& Rulesets**

4. Solution Overview



Great
Ecosy
B

Se

4 NIL

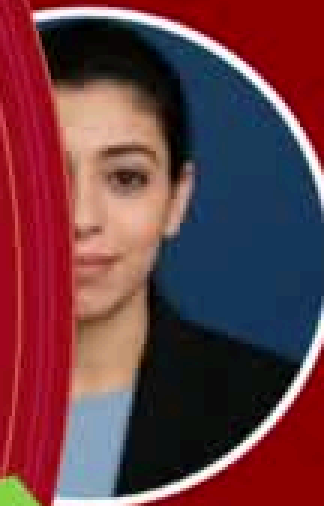
BAL DES PROJETS

12^{ème} édition

2025



Haima Memi



a Dhrief

M

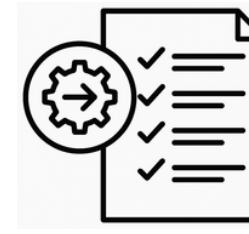


Moslem H

Sinda Galai



**AI-Driven Threat
Detection**



**Automated response
playbooks to reduce
reaction time**

Key Features



**Threat hunting with
behavior analytics &
threat intelligence feeds**

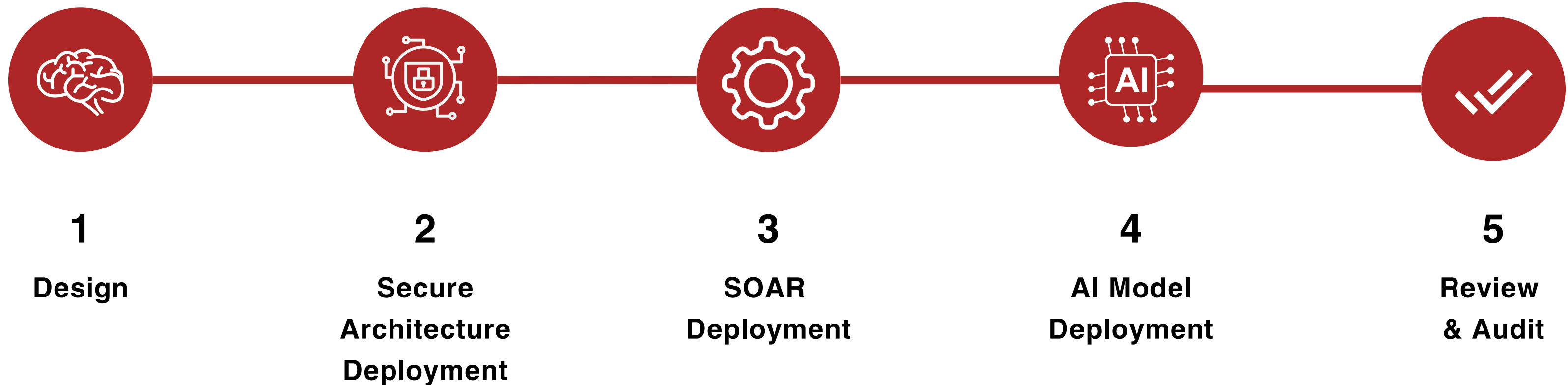


**Compliance-ready audits
& detailed security
reporting**

5. Methodology & Implementation Plan



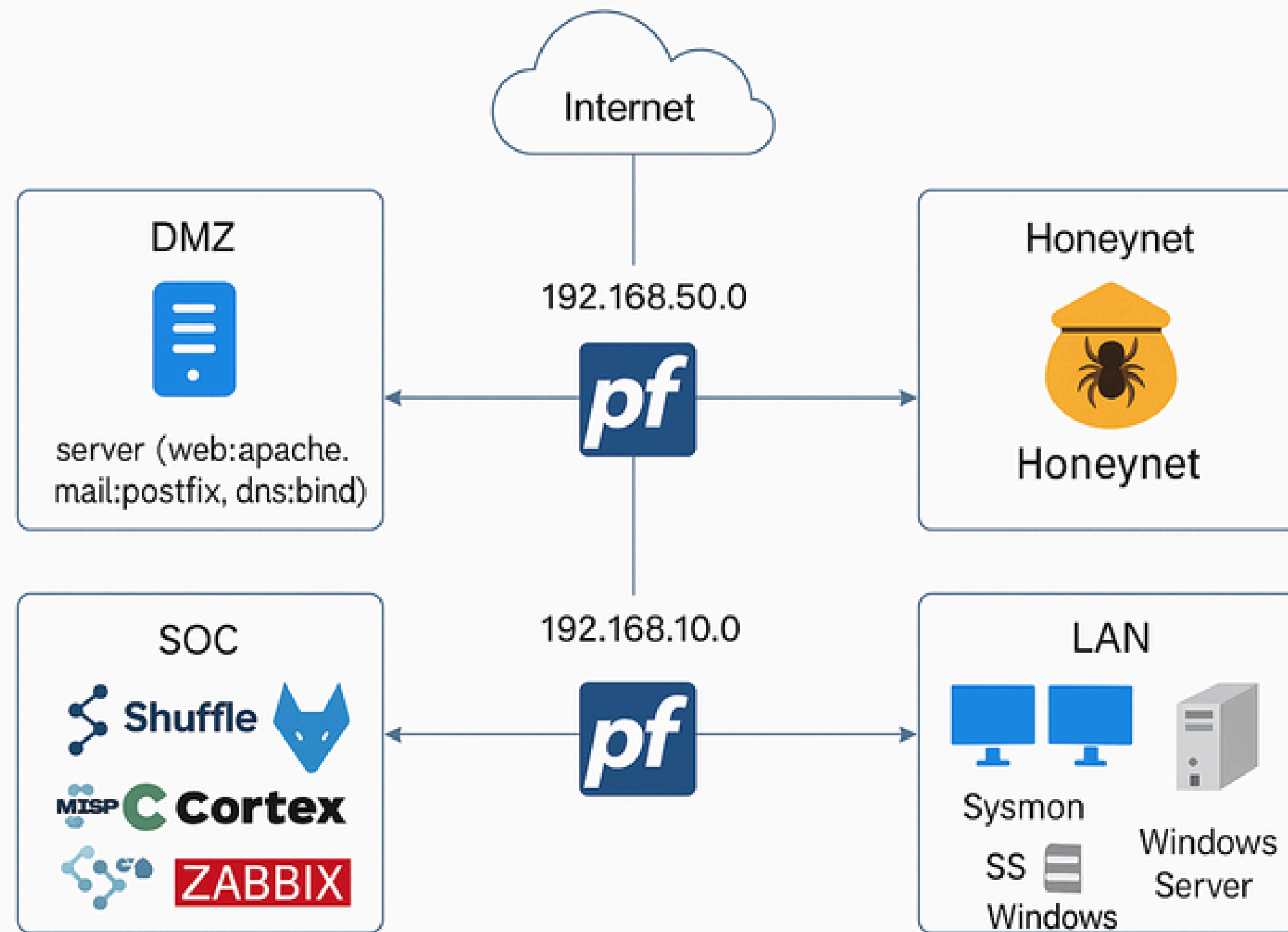
Project phases



6. Architecture & Technical Highlights

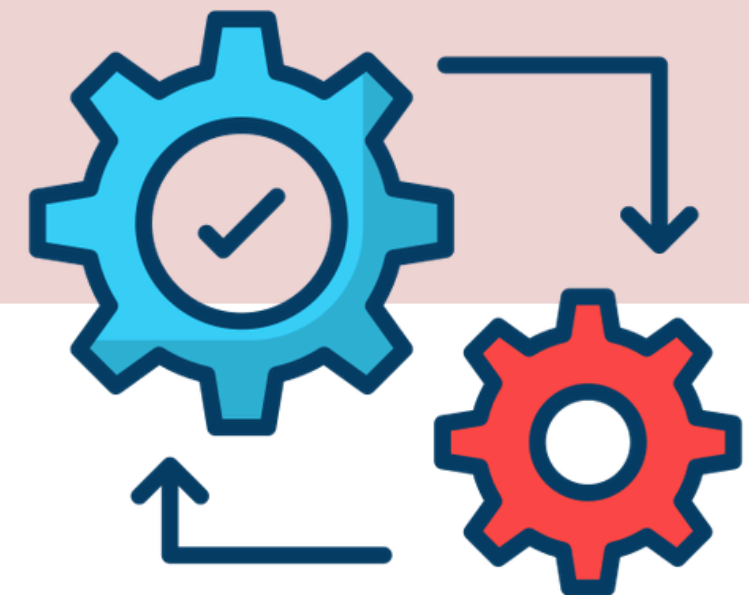


ARCHITECTURE



Data overview & Integration

1. Wazuh collects NDR, EDR, and Zabbix data
2. Events are sent to Shuffle (SOAR)
3. A case is created in TheHive
4. AI model analyzes the data to detect patterns and identify potential threats
5. Analyze by Cortex
6. Enriched with threat intelligence from MISP
7. Automated response triggered via the firewall API



Real-time Capabilities & Scalability

1. Real-time threat detection via Wazuh (EDR, NDR, Zabbix)
2. Events analyzed and orchestrated by Shuffle
3. Cases auto-created in TheHive, enriched by MISP
4. AI model performs real-time threat detection and anomaly analysis
5. Rapid response via firewall APIs
6. Scalable architecture for growing banking needs



7. Resource & Infrastructure Requirements



Infrastructure Requirements

SIEM &
Monitoring

wazuh.

ZABBIX

Threat
Intelligence



HoneyPot



Infrastructure Requirements

SOAR & Automation


Cortex

 **TheHive**



Logging & Endpoint Detection



Web/ Mail/ DNS Services (DMZ)



Apache



Postfix

Team Roles



Ahmed Maalej

- Pfsense
- Honeypot
- AD



Chaima Memi

- Zabbix
- Shuffle
- App Django



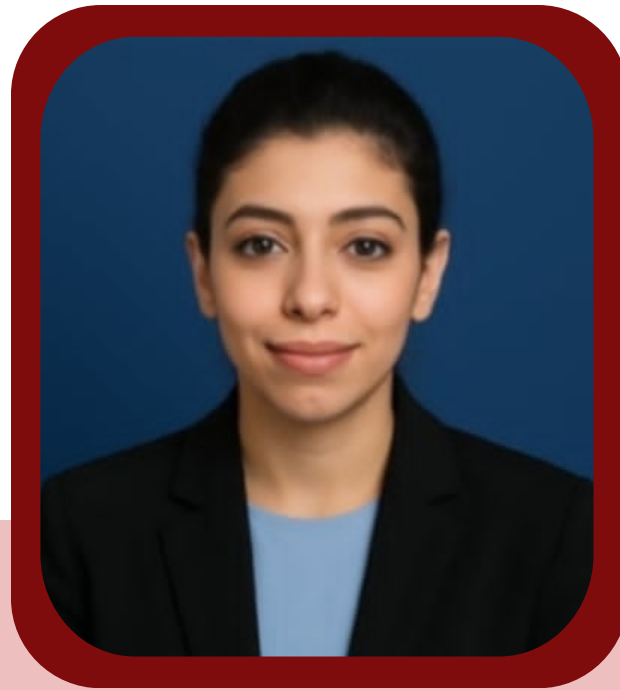
Moslem Haddedi

- Pfsense
- Syslog
- Sysmon



Sinda Galai

- DMZ
- Misp
- Shuffle



Molka Dhrief

- NDR
- Wazuh
- AD



Mohamed Gassoumi

- EDR
- The hive
- Shuffle



Nadhem Laameri

- LAN
- Cortex
- Shuffle

8. Risks, Compliance & Assumptions



RISK & COMPLIANCE



Risks

- Insider threat / Data breach
- Security tool misconfigurations
- Complex integrations
- Zero-day attacks

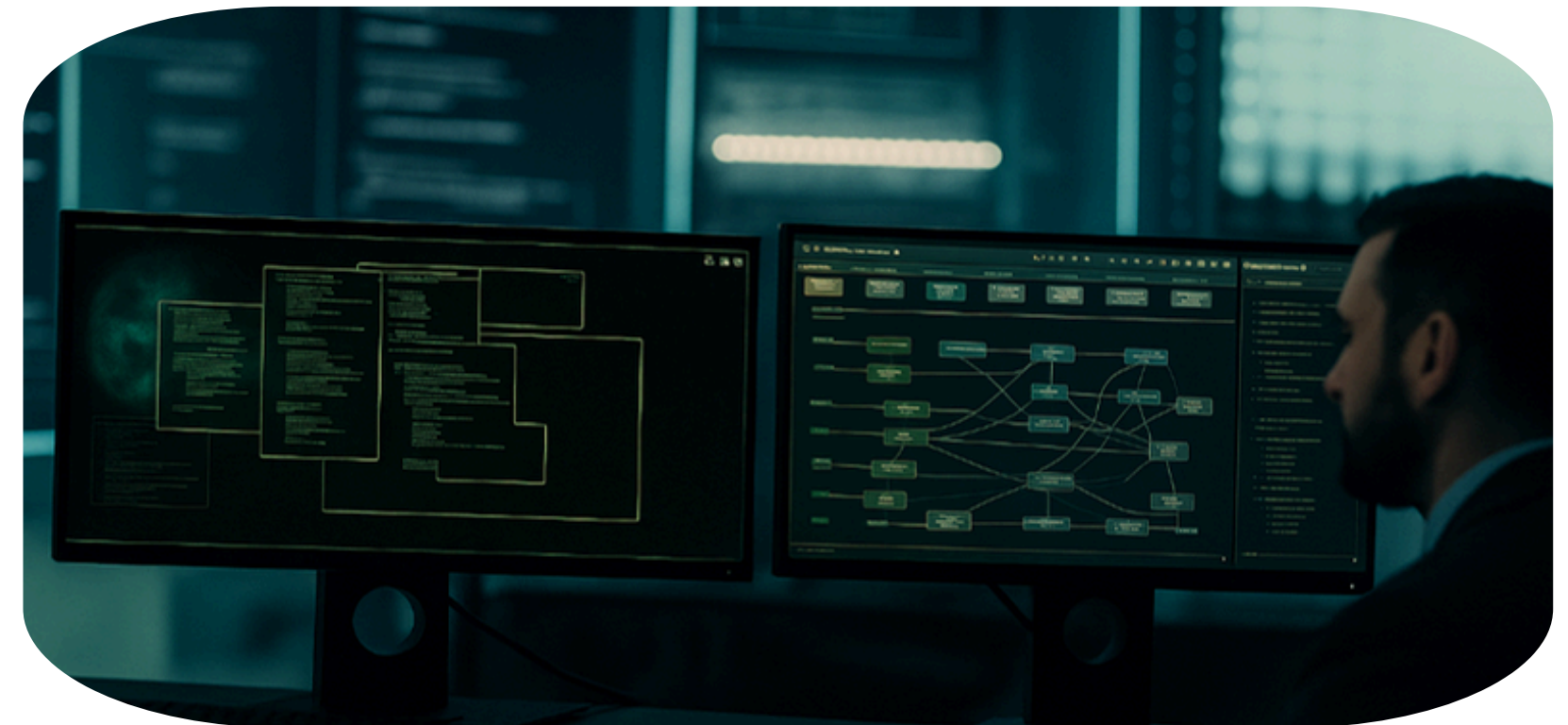


Compliance

- GDPR: Notify incidents < 72h
- Central Bank Guidelines
- Audit-ready reporting

KEY ASSUMPTIONS

- Active and consistent log sources (Sysmon, Syslog)
- Team familiar with open-source tools
- Internet access for updates and threat feeds
- Isolated setup simulates real banking system



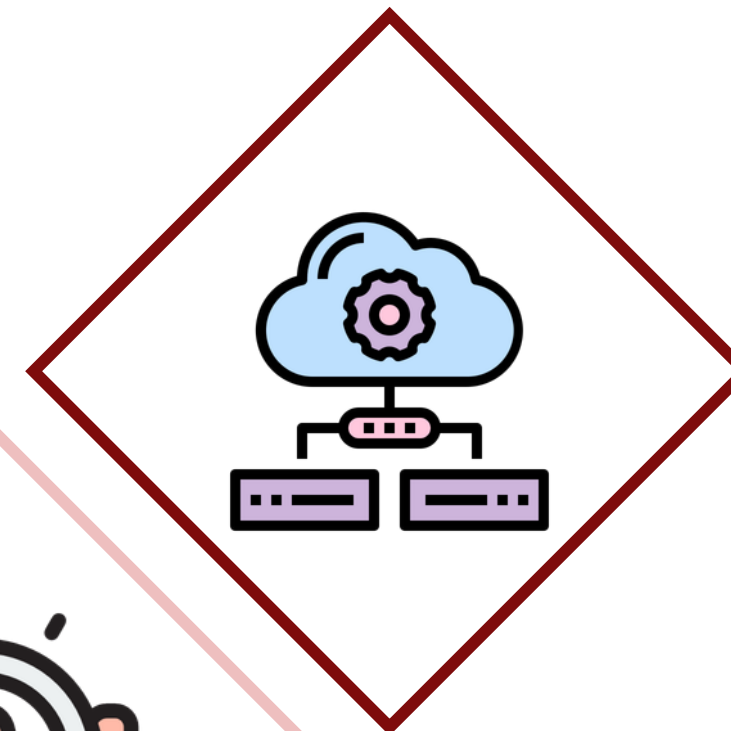
9. Cost Breakdown & ROI



COST BREAKDOWN

LICENSING

- Wazuh SIEM – Open-source
- SOAR (Shuffle, TheHive, Cortex, MISP)
- Firewall



INFRASTRUCTURE

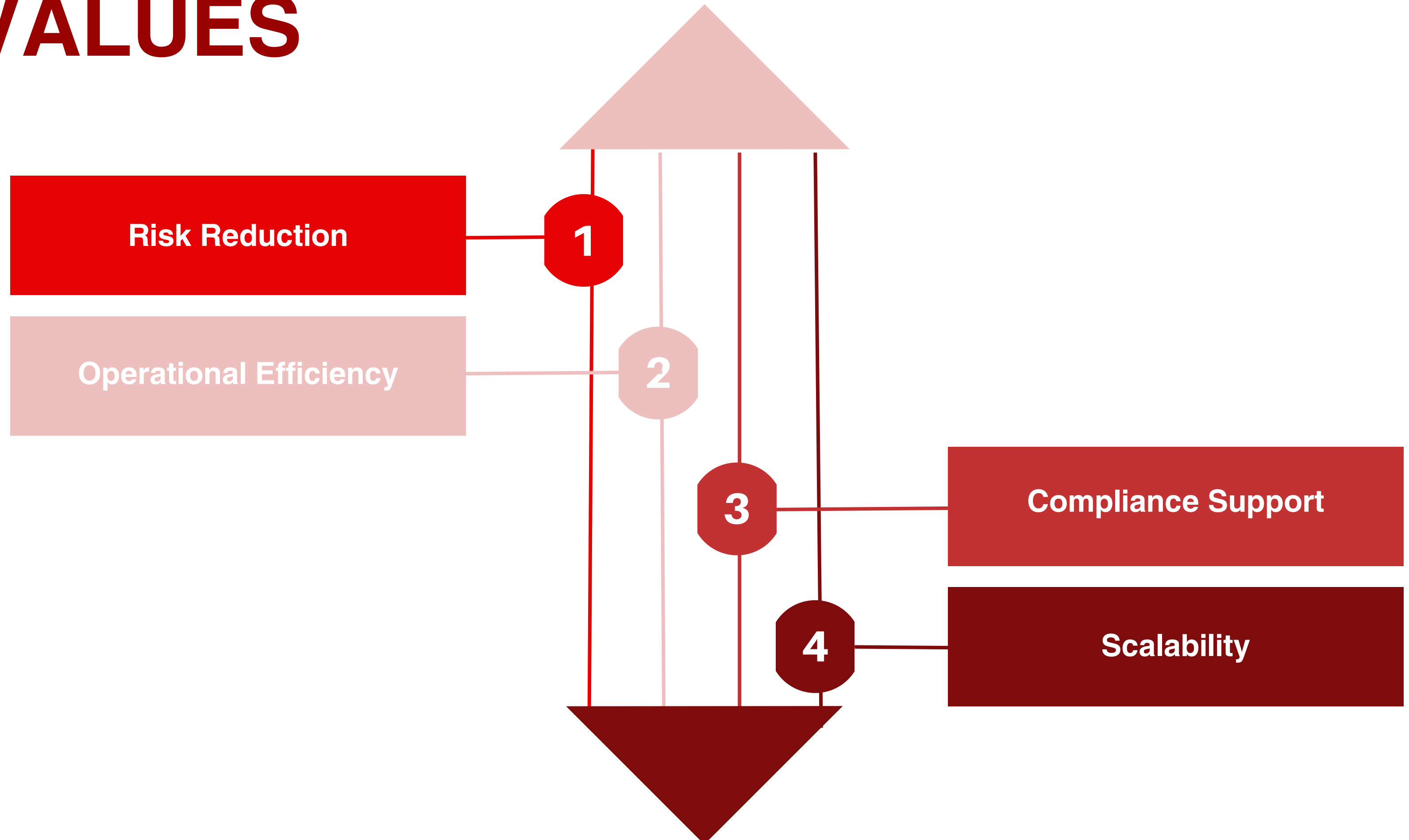
- Servers/VMs for storage & tools
- Network & firewall setup



HUMAN RESOURCES

- Initial setup team
- SOC analysts for ongoing ops & response

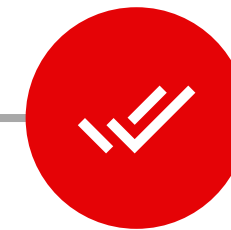
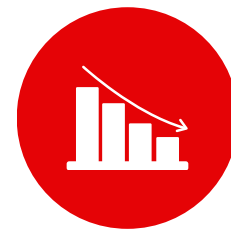
VALUES



ROI

Speed

Achieve up to 60% reduction in incident response time



Cost Savings

30–50% savings on security operations through automation and open-source adoption

Expected ROI

10.KPI & Performance Metrics



SECURITY PERFORMANCE METRICS

- **Detection Accuracy :**
 $\geq 95\%$ with fewer false positives
- **MTTD :**
< 5 min via real-time monitoring
- **MTTR :**
< 10 min with automated response



BUSINESS IMPACT METRICS

- **Incident Reduction**
 - ✓ 70% fewer attacks with early detection & auto-blocking.
- **Efficiency Boost**
 - ✓ 30–50% fewer manual tasks via SOAR automation.
- **High Availability**
 - ✓ >99.9% uptime through enhanced monitoring.



11. Results & Business Value



BUSINESS VALUE

01

Risk Mitigation

Reduces disruptions (e.g., ransomware attacks).

03

Cost Savings

Open-source + less manual work = lower costs.

02

Operational Efficiency

Automates tasks, boosts focus on critical threats.

04

Regulatory Compliance

Supports audits and aligns with standards (PCI, ISO, GDPR).

12. Conclusion & Next Steps



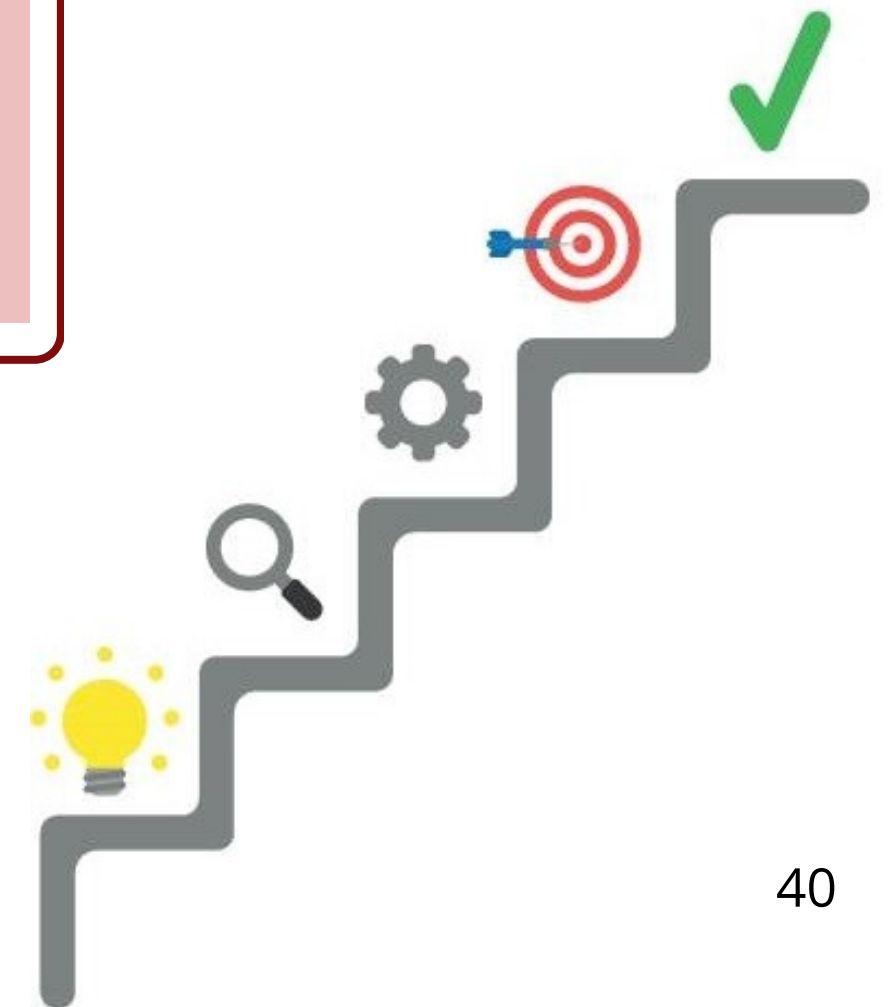
SUMMARY OF OUTCOMES

- ✓ Deployed an AI-driven threat intelligence platform for banking.
- ✓ Enabled real-time detection & automated response.
- ✓ Reduced incidents, improved compliance, and boosted security efficiency.



NEXT STEPS

- Scale deployment across all branches and critical systems.
- Enhance AI models with more data for improved threat prediction.
- Continuous monitoring and auditing to ensure long-term performance.
- Train internal teams for platform operation and incident handling.
- Plan for future integrations (e.g., fraud detection, cloud security).



Thank You!



Social Media
@PiSecurity



Email
PiSecurity@gmail.com



Call us
25 366 650



TEAM
SECURE 7

THREAT INTELLIGENCE ECOSYSTEM FOR BANKING

- Fostering innovation through intelligent automation
- Strengthening digital security for institutions
- Partnering for a safe future

[LinkedIn Link](#)



Maalej Ahmed • Vous
IT engineering student
50 min • Modifié •

🚨 Banking Security Reinvented with AI-Powered Threat Intelligence 🔒

In 2021, I went to a bank for a simple transaction—only to find the IT system completely down due to a ransomware attack. That disruption inspired a bold idea: What if we could prevent this from ever happening again?

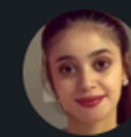
That moment led to the creation of our automated Threat Intelligence Platform—custom-built for banking environments.

🔍 What it does:

We combined Wazuh (SIEM) with real-time data from EDR, NDR, and Zabbix, orchestrated through Shuffle (SOAR).

Cases are managed in TheHive, analyzed with Cortex, enriched via MISP, and reinforced by an AI model that detects threats in real time, triggering automated firewall responses.

🏠 The result? Faster detection, smarter response, and stronger



Chaima Mami • 1er
Étudiante à esprit
32 min •

🚨 Banking Security Reinvented with AI-Powered Threat Intelligence 🔒

In 2021, I went to a bank for a simple transaction—only to find the IT system completely down due to a ransomware attack. That disruption inspired a bold idea: What if we could prevent this from ever happening again?

That moment led to the creation of our automated Threat Intelligence Platform—custom-built for banking environments.

🔍 What it does:

We combined Wazuh (SIEM) with real-time data from EDR, NDR, and Zabbix, orchestrated through Shuffle (SOAR).

Cases are managed in TheHive, analyzed with Cortex, enriched via MISP,



Molka Dhf • 1er
Étudiant à Ecole Supérieure Privée d'Ingénierie et de Technologies...
15 min • Modifié •

🚨 Banking Security Reinvented with AI-Powered Threat Intelligence 🔒

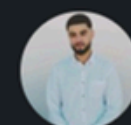
In 2021, I went to a bank for a simple transaction—only to find the IT system completely down due to a ransomware attack. That disruption inspired a bold idea: What if we could prevent this from ever happening again?

That moment led to the creation of our automated Threat Intelligence Platform—custom-built for banking environments.

🔍 What it does:

We combined Wazuh (SIEM) with real-time data from EDR, NDR, and Zabbix, orchestrated through Shuffle (SOAR).

Cases are managed in TheHive, analyzed with Cortex, enriched via MISP,



Mohamed Gassoumi • 1er
Étudiant à l'Ecole Supérieure Privée d'Ingénierie et de Technologies...
23 min • Modifié •

🚨 Banking Security Reinvented with AI-Powered Threat Intelligence 🔒

In 2021, I went to a bank for a simple transaction—only to find the IT system completely down due to a ransomware attack. That disruption inspired a bold idea: What if we could prevent this from ever happening again?

That moment led to the creation of our automated Threat Intelligence Platform—custom-built for banking environments.

🔍 What it does:

We combined Wazuh (SIEM) with real-time data from EDR, NDR, and Zabbix, orchestrated through Shuffle (SOAR).

Cases are managed in TheHive, analyzed with Cortex, enriched via MISP, and reinforced by an AI model that detects threats in real time, triggering automated firewall responses.

 **Sinda Galai** • 1er
Computer Engineering Student
17 min • Modifié • 

🚨 Reinventing Banking Security with AI-Driven Threat Intelligence 🧠🔒

Back in 2021, I witnessed a bank's IT system crash from a ransomware attack. That moment sparked an idea: What if we could stop threats before they hit?

💡 Fast forward to today: we built an AI-powered Threat Intelligence Platform tailored for banking, integrating Wazuh (SIEM), EDR, NDR, Zabbix, TheHive, Cortex, MISP & Shuffle (SOAR) for real-time detection and automated response.

🔒 Results? Faster detection. Smarter response. Stronger protection.

Built through 5 key phases:

- 1 Design
- 2 Secure Deployment
- 3 SOAR Automation
- 4 AI-Driven Threat Intelligence
- 5 Review & Pentest

👏 Huge thanks to my amazing team: [Chaima Mami](#), [Molka Dhf](#), [Moslem Haddadi](#), [Maalel Ahmed](#), [Mohamed Gassoumi](#), [Nadhem](#)

 **Nadhem Laamiri** • 1er
Étudiant en cycle d'ingénieur en IT
7 min • Modifié • 

🚨 Banking Security Reinvented with AI-Powered Threat Intelligence 🧠🔒

In 2021, I went to a bank for a simple transaction—only to find the IT system completely down due to a ransomware attack. That disruption inspired a bold idea: What if we could prevent this from ever happening again?

That moment led to the creation of our automated Threat Intelligence Platform—custom-built for banking environments.

🔍 What it does:
We combined Wazuh (SIEM) with real-time data from EDR, NDR, and Zabbix, orchestrated through Shuffle (SOAR).

Cases are managed in TheHive, analyzed with Cortex, enriched via MISP, and reinforced by an AI model that detects threats in real time, triggering automated firewall responses.

🏠 The result? Faster detection, smarter response, and stronger

 **Moslem Haddadi** • 1er
ESPRIT (Ecole Supérieure Privée d'Ingénierie et de Techno...
6 h • 

🚨 Banking Security Reinvented with AI-Powered Threat Intelligence 🧠🔒

In 2021, I went to a bank for a simple transaction—only to find the IT system completely down due to a ransomware attack. That disruption inspired a bold idea: What if we could prevent this from ever happening again?

That moment led to the creation of our automated Threat Intelligence Platform—custom-built for banking environments.

🔍 What it does:
We combined Wazuh (SIEM) with real-time data from EDR, NDR, and Zabbix, orchestrated through Shuffle (SOAR).